

AI Usage

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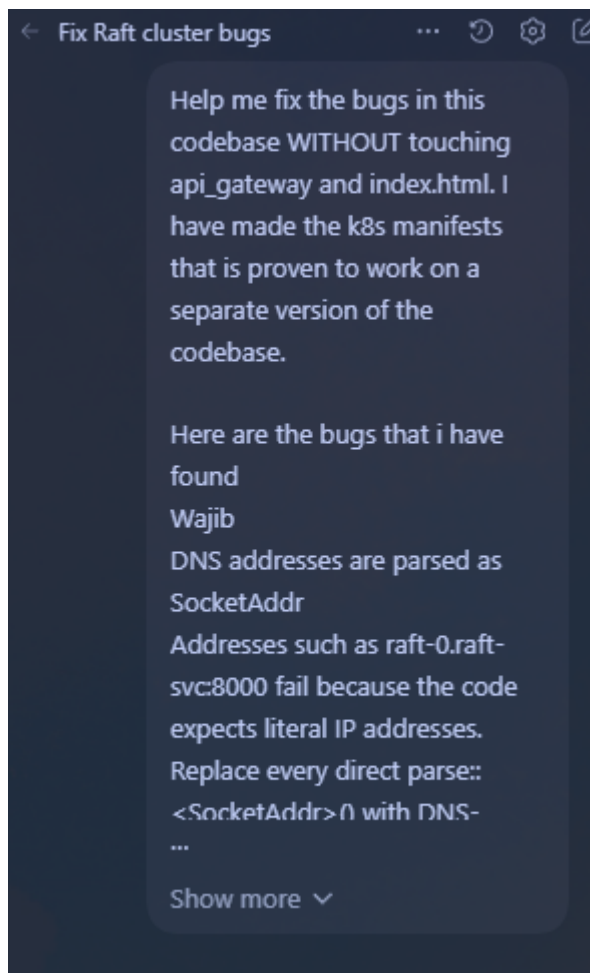
-ChatGPT

Untuk debugging dan troubleshooting, membantu dalam penggunaan dan penulisan commands kubernetes, penulisan [README.md](#), membantu memperbaiki penulisan manifests kubernetes membantu mempelajari konsep yang diperlukan. Detail chat berada di link di bawah :

<https://chatgpt.com/share/6a70c45c-27bc-83ec-9cf9-34f1db865549>

-Codex

Untuk membantu memperbaiki bugs yang ada di codebase Rust.



Full prompt “

Help me fix the bugs in this codebase WITHOUT touching `api_gateway` and `index.html`. I have made the k8s manifests that is proven to work on a separate version of the codebase.

Here are the bugs that i have found

Wajib

DNS addresses are parsed as `SocketAddr`

Addresses such as `raft-0.raft-svc:8000` fail because the code expects literal IP addresses.

Replace every direct `parse::<SocketAddr>()` with DNS-aware resolution using `tokio::net::lookup_host()`.

`execute_with_redirect()` is incompatible with the original API Gateway

It expects `SocketAddr`, while the Gateway provides a `String`.

Make the shared client function accept hostname strings.

Leave `api_gateway.rs` unchanged.

Nodes connect to peers only once

Early Pods cannot connect to peers that have not started yet.

Add periodic reconnection for missing peers, including before elections.

Each node includes itself inside the peer map

This causes self-connections and incorrect quorum calculations.

Remove the local node from peers and store its address separately as `my_addr`.

Leader election blocks the actor

`start_election()` waits for vote responses and prevents the actor from processing incoming vote or heartbeat requests.

Send vote requests asynchronously and return results through actor messages.

Election timer fires immediately

The first `tokio::time::interval()` tick happens immediately, causing simultaneous elections.

Use a randomized Sleep or deadline timer that waits before its first election.

Advertised addresses use localhost and incorrect ports

Addresses such as 127.0.0.1:8001 cannot be reached by other Pods.
Advertise stable StatefulSet DNS addresses such as raft-0.raft-svc:8000.
PVC storage is mounted but unused
State files are written under /app, rather than /data.
Set the Raft container working directory to /data, or use absolute persistence paths.
Restart recovery is incomplete
commit_index is not persisted.
Committed log entries after the latest snapshot are not replayed.
Installed snapshots are not saved to disk.
Fix all three so deleted Pods recover their actual key-value state.
Persistence errors are ignored
Several operations use let _ = persist_state().
Votes, log replication, snapshots, and writes must not return success when persistence fails.
Readiness only checks whether port 8000 is open
A node can appear Ready while no leader exists.
Add an application-aware readiness mechanism that checks recovery and cluster participation.
Gateway depends specifically on raft-0
Deleting raft-0 temporarily breaks Gateway access.
Use the headless Service as the initial contact and let the client try multiple resolved addresses.
Scaling 3 to 5
New StatefulSet Pods do not execute AddNode
raft-3 and raft-4 start independently.
New ordinals must bootstrap through an existing node and request membership addition.
Membership changes exist only in the leader's memory
Followers do not update their actual Raft peer configuration when applying AddNode or RemoveNode.
Membership changes must be applied by every node from committed log entries.

Membership changes are acknowledged before commitment

AddNode and RemoveNode return success immediately after appending.

Return success only after the membership entry is replicated, committed, and applied.

Membership is not persisted

Restarting nodes restores the original static ConfigMap membership.

Persist the committed membership and rebuild peer clients during recovery.

Joining nodes do not synchronize existing data

New nodes need log catch-up or snapshot installation before becoming Ready.

Chaos bonus

Pending requests are not cleared after leadership loss

Gateway requests can hang when the leader steps down.

Fail or redirect pending requests when leadership changes.

Stale RPC results can modify a newer leader term

AppendEntries responses should include the originating term.

Ignore responses from previous terms or previous leadership periods.

Leader address can remain stale

Clear current_leader when starting an election or observing a higher term.

NetworkPolicy and RBAC bonus

NetworkPolicy does not permit DNS

DNS-aware peer connections will fail once egress isolation is active.

Allow TCP and UDP port 53 to CoreDNS.

NetworkPolicy does not fully isolate the Gateway

Add default-deny policies plus explicit Gateway-to-Raft, Raft-to-Raft, DNS, and Gateway ingress rules.

ServiceAccounts must match workload manifests

StatefulSet should use raft-sa.

Gateway should use gateway-sa.

The empty Role is acceptable because neither application accesses the Kubernetes API directly.”

Worked for 12m 29s >

Implemented the Raft/Kubernetes fixes without touching `api_gateway.rs` or `index.html`.

Key changes:

- DNS-aware RPC connections via `tokio::net::lookup_host`, including redirect and headless-service retries.
- Stable advertised StatefulSet DNS addresses; `/data` is now the working directory.
- Removed self from peer membership and persisted `my_addr` separately.
- Non-blocking election vote handling with actor messages and randomized `Sleep` timer.
- Periodic peer reconnection through heartbeat/replication paths.
- Persisted commit index and committed membership; `reconcile` replays post-snapshot committed logs.

- Installed snapshots are atomically saved to disk.
- Membership changes apply on all nodes and only acknowledge after commit/application.
- Pending client/membership requests fail on leadership loss; stale replication results are ignored.
- Application-aware Raft readiness via `/data/raft.ready`.
- Gateway uses `raft-svc:8000`, avoiding a dependency on `raft-0`.
- StatefulSet shell startup fixed for `/bin/sh`; NetworkPolicies now default-deny with explicit Raft, Gateway, and CoreDNS traffic.

Validation completed:

Validation completed:

text



cargo test

All tests pass.



Edited...

+374 -25

Undo ↶

Review

src/raft/state.rs

+35 -17

src/utils/client.rs

+45 -15

src/bin/server.rs

+14 -19

Show 6 more files ▼

The UI shows Error: Invalid
RAFT_ADDR: raft-svc:8000

Worked for 35s >

This is a stale-container issue, not the current
source: `api_gateway.rs` now passes
`RAFT_ADDR` as a `String`, and the text
`Invalid RAFT_ADDR` does not exist in this
checkout.

Your Gateway Pods are still running an older
`hogyoku:dev` image, likely retained because
the deployment uses:

yaml



```
imagePullPolicy: IfNotPresent
```

Rebuild/load the image into your Kubernetes
cluster, then restart the deployment:

sh



```
docker build -t hogyoku:dev .  
kubectl -n hogyoku rollout restart  
kubectl -n hogyoku rollout restart  
kubectl -n hogyoku rollout status
```

If using Minikube, load it first:

sh



```
minikube image load hogyoku:dev
```

The manifest's `RAFT_ADDR: raft-svc:8000`
is correct for the updated DNS-aware client.

